

Finite scalar matching, higher determinant phases and sequestering

Exact Gaussian poles and residues, a neutral-scalar protection
criterion and correlated wall metrics

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Finite UV matching and reproducible phase counterexamples with explicit loop-order scope.
Includes an exact bridge to the merged conditional cosmology tools.

The finite matching result

Gaussian scalar mediation correlates a potential, a derivative expansion and finite pole normalization. This chapter derives those correlations exactly and evaluates them on the existing nonet/Higgs branch. All 21 light scalar poles obtained from the momentum-dependent source operator agree with the full 48- and 76-coordinate UV matrices to approximately 5 times 10 to the minus 14 in squared-mass units. This closes the finite tree scalar pole and residue problem for the specified Gaussian completions, including unequal positive mediator masses in the general implementation. It does not identify source-field normalization with quark normalization or derive a mixing angle.

The second result is a precise boundary on the mass-phase theorem. First-order phase cancellation remains intact. An exact rational spectral-kernel example and an independently evaluated finite B0 example show that the quadratic term in the determinant of the one-loop-corrected mass matrix can nevertheless be nonzero. Separating Higgs and source scalar modes protects that restricted resummed determinant, but the specified nonzero Higgs and source Yukawas regenerate Higgs portals at one loop. Thus this particular protection mechanism has an exact renormalization test, rather than an assumed extension of first-order reality.

The Gaussian derivative metric also connects directly to the other session's cosmology tools. Its latest merged calculation lifts the actual metastable-decay trial paths to all 48 canonical fields; the pole and residue results here give the corresponding finite-spectrum description. A rigorous wall-tension sandwich and an exactly classified two-field example are implemented in both SymPy and the existing BigInt polynomial-path engine. The example has genuine global minima; no cosmological tension is assigned to the retained quark-flavor branch, which the preceding chapter proved is not globally selected by its action.

Exact scalar poles and canonical residues

$$F(z) = H - zK(z), \quad Z(z) = -F'(z)$$

On the stationary Gaussian valley $S_a = -J_a/M_a$ squared, write H for the source/Higgs Hessian and $D_{ai} = \text{partial}_i J_a$. The full canonical UV Hessian has blocks $H + D\text{-transpose diag}(1/M_a \text{ squared}) D$, $D\text{-transpose}$, D , and $\text{diag}(M_a \text{ squared})$. Its static Schur complement is exactly H , so its stationary energies and the signs of its light static curvatures reproduce the original source potential. Its finite pole masses are different because heavy fields carry canonical kinetic terms.

For a timelike squared momentum z below every mediator threshold, eliminating the heavy fluctuation gives $F(z) = H - zK(z)$, where $K(z) = I + D\text{-transpose diag}[1/(M_a \text{ squared} (M_a \text{ squared} - z))] D$. The full characteristic determinant factors into $\det[\text{diag}(M_a \text{ squared}) - z I] \det F(z)$. Exact symbolic checks establish the one-source identity for general h, d, M, z and a polynomial identity with two source coordinates and three unequal mediators. The general formula follows directly from the same block Schur complement. No field-to-heavy-mass expansion is used for the pole equation.

The two-derivative tree metric is $K(0) = I + D\text{-transpose diag}(1/M_a \text{ to the fourth}) D$. At a root z , the physical residue metric is $Z(z) = -\text{partial}_z F(z) = I + D\text{-transpose diag}[1/(M_a \text{ squared} - z) \text{ squared}] D$. If $F(z)v = 0$, its heavy component is $-\text{diag}[1/(M_a \text{ squared} - z)] Dv$. The full vector is canonically normalized precisely when $v\text{-transpose } Z(z) v = 1$. Distinct full eigenvectors are orthogonal even though their projected source vectors use different pole metrics. Degenerate source modes are handled by the full nullspace basis; individual eigenvectors inside a degenerate subspace have no invariant identity.

These formulas distinguish a zero-momentum curvature, a local derivative approximation and a finite UV pole. For a positive source Hessian whose largest eigenvalue is below the smallest mediator squared mass, every ordered root obeys $\text{lower_bound}_i \leq \text{exact_pole}_i \leq \text{two_derivative_pole}_i \leq \text{bare_curvature}_i$. A useful lower bound divides the ordered eigenvalue of H by $1 + \text{norm}(D)^2 / [M_{\min}^2 (M_{\min}^2 - \lambda_{\max}(H))]$. Here $\text{norm}(D)$ is the spectral norm. It follows from the positive matrix bound on $K(z)$ and the min-max principle. The momentum-dependent eigenvalues of $F(z)$ decrease strictly because $-\partial_z F$ is positive; each branch therefore has one root in the stated below-threshold interval. The numerical solver independently checks the resulting roots, full canonical vectors and collective light-subspace projector against the full matrix.

At the existing finite-only completion, the tree metric's largest eigenvalue is 1.0011338172 and its up/down source kinetic block has spectral norm 0.0003900162. The local two-derivative approximation's maximum relative pole error is 0.0000666455. For the universal completion the corresponding numbers are 1.0012836495, 0.0004148295 and 0.0000781621. The full finite poles agree to 4.98 times 10 to the minus 14 and 2.85 times 10 to the minus 14 respectively. These small errors support this derivative expansion at the declared mediator mass 10; they are not bounds for arbitrary masses or changed coefficients.

In a mostly-minus Minkowski convention, integrating out canonical Gaussian scalars gives the Lagrangian contribution $+J\text{-transpose} (M^2 + \text{Box})\text{-inverse} J/2$. Its zero-derivative effective potential is $-J^2/(2M^2)$; integration by parts gives the positive kinetic term $(\partial J)^2/(2M^4)$. The higher derivative terms are fixed by the same inverse operator. Potential coefficients, scalar derivative coefficients and scalar residues therefore cannot be chosen independently within this completion. The former tree matching of a quartic potential is recovered as the zero-derivative limit. The new tree derivative term is distinct from the preceding chapter's one-loop wavefunction threshold.

What first-order mass-phase reality does and does not enforce

For a canonical triangular fermion matrix $D = (aI, 0; C, M I)$, C Hermitian and a, M real positive, each neutral real-scalar vertex has the form $G_\alpha = (b_\alpha I, 0; H_\alpha, 0)$, with b_α real and H_α Hermitian. The established paired-vertex theorem gives $\text{Im Tr}(D\text{-inverse} \delta D) = 0$ for arbitrary real scalar mixing and positive scalar masses. This remains a first-order statement in the one-loop mass correction.

Every corrected vertex contribution still has zero right column block, so the determinant of $D + \delta D$ depends only on its corrected upper-left three-by-three block. In the basis diagonalizing C , a real spectral kernel $f_\alpha(D D^\dagger)$ has upper contractions $L_\alpha(C)$ and $K_\alpha(C)$, both real functions of C . The upper mass correction is proportional to the sum of $b_\alpha^2 L_\alpha(C) + b_\alpha K_\alpha(C) H_\alpha$. The second term need not be Hermitian because its two Hermitian factors need not commute. Its trace is real; its higher trace powers need not be.

An exact resolvent kernel makes this distinction algebraically transparent. For $f_s(X) = (X + sI)\text{-inverse}$, $L_s = a(M^2 + s)/\text{den}$ and $K_s = sC/\text{den}$, with $\text{den} = (a^2 + s)(M^2 + s) + sC^2$. Take $C = \text{diag}(2/5, 7/10, 11/10)$, $a = 1/10$, $M = 1$ and scalar vertices consisting of a Higgs coefficient $1/5$, a real 12 off-diagonal source coefficient $1/5$ and an imaginary Hermitian 12 source coefficient $1/5$. Mix their three scalar fields with the rational orthogonal matrix having rows $(1, 2, 2)/3$, $(2, 1, -2)/3$ and $(2, -2, 1)/3$; the squared masses are $3/10, 4/5, 7/5$. The linear determinant phase coefficient vanishes exactly. The quadratic

coefficient is the nonzero rational $-5152/727803505$ before the loop-normalization factor. Exact expansion of the full three-by-three determinant verifies the coefficient.

The resolvent is a structural example, not a substitute for the physical loop function. Evaluating the actual finite B0 function for the same data gives a quadratic determinant phase -4.480088664908940366 times 10 to the minus 10 radians and a resummed phase -4.481809518794191284 times 10 to the minus 10 radians. The full six-by-six SVD calculation agrees with an independent 80-digit divided-difference evaluation of the upper mass block. Its relative correction norm is below 0.000533 ; its first-order phase is zero analytically. Formal scaling of the one-loop correction verifies the quadratic approach to the zero-correction limit.

The independent four-scalar example imposes an additional structural restriction: the Higgs Hessian row couples only to a diagonal source, with zero direct Higgs mixing into either off-diagonal source. The diagonal source mixes with the real and imaginary off-diagonal sources in the remaining positive Hessian. The first-order phase is below 10 to the minus 16 , while the resummed one-loop matrix phase is approximately 1.4070541 times 10 to the minus 10 with relative correction norm below 0.02 . This shows that a Higgs row aligned to a commuting source direction is not alone enough to extend the first-order theorem through propagation across the scalar sector.

Both examples are spectral tests. A complete invariant scalar potential with those Hessians, its stationary vacuum and all gauge/fermion sectors have not been supplied. More importantly, the quadratic determinant term of a one-loop-corrected matrix is only one contribution at the next perturbative order. Genuine two-loop self energies and the remaining matching terms can contribute at that order. These examples disprove the inference that first-order reality automatically protects the resummed determinant; they do not calculate a complete two-loop strong-CP prediction for the repository model.

A protective candidate with an exact running test

If no scalar mass eigenmode contains both a Higgs vertex and a source vertex, source modes have $b_{\alpha} = 0$ and do not correct the top block. Higgs modes have $H_{\alpha} = 0$ and correct it only by real functions $L_{\alpha}(C)$. The corrected top block is then Hermitian and commutes with C . Its determinant ratio remains real and positive whenever the normalized correction norm is below one. This protects every power in that restricted determinant resummation. It remains a statement about one-loop matrices; it does not address all diagrams at arbitrary loop order.

The candidate separation is not stable under the stated Yukawa interactions. At zero Higgs portals, pure source quartics and the Higgs self quartic share no scalar-loop product, while off-diagonal source wavefunctions cannot generate a source/Higgs product from those separated potentials. The colored fermion box does generate them. For each real charge sector with Higgs coefficient y , source singlet coefficients e, t and adjoint coefficient a , its four source-quadratic portal beta coefficients are $-36 y^2 e^2$, $-36 y^2 t^2$, $-72 y^2 e t$ and $-12 y^2 a^2$. These are $16 \pi^2$ -normalized coefficients multiplying $H^\dagger H$ times η^2 , s^2 , ηs and the adjoint norm respectively.

All four vanish exactly only if $y = 0$ or $e = t = a = 0$ in that sector. The diagonal coefficients establish this directly by positivity for real couplings. Nonzero Higgs masses together with nonzero source mixing therefore prevent this zero-portal subspace from closing under the specified running. Ordinary phase charges cannot forbid these neutral products while allowing their factors. Additional particles, a different interaction structure or justified cancellation relations can alter the conclusion; none is supplied by setting these portals to zero as a boundary input. The new result specifies exactly what a protection mechanism must change.

Connecting the scalar metric to conditional CP walls

$$\sigma_{\text{EFT}} \leq \sigma_{\text{UV}} \leq \sigma_{\text{valley}}$$

The merged cosmology layer supplies exact polynomial-path bounds for canonical full fields. Gaussian matching provides the previously missing derivative contribution on its valley: for any path $x(t)$, the canonical speed squared is $\text{norm}(x\text{-prime})^2 + \sum_a (J_a\text{-prime})^2/M_a^2$ to the fourth. This is precisely the induced tree metric $K(0)$, evaluated on the path. A source-only canonical path omits that positive term.

When the supplied endpoints are degenerate global minima of the source potential and the Gaussian completion, every full static profile has energy at least that of its projected canonical source profile: the heavy-field kinetic energy and the Gaussian potential square are nonnegative. Conversely, the Gaussian valley is an admissible full-field path. Taking infima gives $\sigma_{\text{canonical_EFT}} \leq \sigma_{\text{full_UV}} \leq \sigma_{\text{valley}}$. The last inequality is an upper-bound construction, not an assertion that a true wall stays on the valley; heavy gradients can favor departure. The endpoints must be global. The quark benchmark's lower CP-conserving competitor prevents applying this stable-wall conclusion to its retained CP-breaking branch.

The exact two-field example is $V = \lambda(x^2 - v^2)^2/4 + M^2(S + g x^2/M^2)^2/2$ with canonical x, S . Its only global minima are $x = \pm v$, $S = -g v^2/M^2$, from simultaneous vanishing of the two positive squares. CP acts as $x \rightarrow -x$ and $S \rightarrow S$. The canonical source wall has tension squared $8 \lambda v^6$. On the polynomial valley path $x = v(2t-1)$, the cosmology engine's Cauchy bound is $\sigma^2 \leq 16 \lambda v^6$ times $[1 + 4 g^2 v^2/(3 M^4)]$.

For $\lambda = v = g = 1$ and $M = 3$, the full UV tension obeys $8/9 \leq \sigma_{\text{full}}^2 \leq 3952/3645$. SymPy integration and the independently implemented BigInt path engine return the same rational upper bound. The latter receives both canonical path coordinates, including $S(t)$, rather than an unexplained metric input. The potential's composition and global endpoints are verified explicitly for this example. These are dimensionless toy results, not a flavor wall tension or thermal-history prediction. The other session's conditional bias optimization and restoration tools remain applicable only when their physical hypotheses are supplied.

Reproducibility and mechanism consequence

Run `PYTHONPATH=python OPENBLAS_NUM_THREADS=1 OMP_NUM_THREADS=1 python python/develop_flavor_spectral_matching.py` to regenerate `receipts/m22_interactions/flavor_spectral_matching.json`. Run `node research/flavor_cosmology/gaussian_wall_bridge.mjs` for the independent rational bridge, and `node research/flavor_cosmology/test_cp_wall_tools.mjs` for the merged cosmology module's 104 assertions. The dedicated Python tests independently check unequal-mass Schur identities, finite pole bounds, residue derivatives, degenerate modes, the full completion spectra, high-precision B_0 phases, separated scalar modes and regenerated portals. Validation records full related-test counts, receipt hashes and fresh-extraction results separately.

The new mechanism obligation is specific: a proposed separation must survive the portal operators forced by its own quark couplings, or supply another exact structure that constrains the higher determinant terms. Gaussian scalar exchange already enforces derivative correlations and permits controlled finite matching, but that scalar relation does not itself fix quark kinetic operators, cross-sector flavor coefficients, golden CKM or a gauge boundary. The first-order phase theorem, new finite pole formulas and new sequestering exclusion coexist without promoting a partial determinant expansion to a full two-loop prediction.

The finite B0 convention and neutral-scalar mass-threshold normalization are inherited from `python/perfectpower/flavor_quantum.py` and the preceding Hermitian-source chapters. The general-RGE primary reference remains <https://arxiv.org/pdf/1809.06797>. Conditional wall mechanics and the literature context are recorded in `docs/FLAVOR_COSMOLOGY_MONOGRAPH.md`. All new spectral and exact rational calculations here are executable repository results; no independent publication-priority review is asserted.